TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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Abstract— **Modern stereo estimators are applied independently to each frame and can fluctuate under specularities, weak texture, tissue deformation, tool occlusions and camera motion. Video stereo methods fix this by building temporal reasoning into the estimator, so a new backbone means a new temporal model. We ask the opposite question: can one causal temporal module be trained once and attached, unchanged, to frozen and heterogeneous stereo estimators?**

We present TETHER, a causal plug-and-play refiner that answers it. One frozen $154{,}874$-parameter checkpoint, trained on three estimators and one dataset, improves *every* estimator we evaluate on *every* arena we evaluate: five architectures — two of them excluded from its training — across development, held-out and out-of-domain splits, all fifteen backbone–arena cells in mean EPE, with the sequence-level exceptions named where they occur. Zero-shot on DRENDS, a dataset no part of the module has seen, disparity EPE, Bad3 and metric depth MAE and RMSE improve in all 25 recording–backbone cells, by -5.40, -19.51, -4.02 and -3.47%, with nothing adapted, retrained or tuned for the domain. The two unseen estimators we cannot separate from the three seen ones, which is the claim.

The module reads only what a black box emits: preceding disparity motion-aligned with frozen SEA-RAFT, and a shared 38-channel evidence space of disparity, motion and image geometry — no learned appearance features, no frozen backbone, no future frame — feeding a reset-and-fusion head that predicts a bounded interpolation between the current estimate and that aligned memory. Under a matched state machine the gain is not temporal smoothing: 3.69% mean EPE against 0.59% for the best fixed or EMA blend, which buys its mean error with bad pixels while the learned gate does not. Accumulated into a map the refined geometry reduces world-frame disagreement in all 20 held-out cells and increases it on development; we identify the term that decides which. We additionally introduce a refinement-centric protocol scoring introduced error alongside recovered error under fixed, prediction-independent support, and apply it to our own claims.

I. Introduction

Metric three-dimensional perception underpins camera navigation, tool–tissue distance estimation and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth comes from images the platform already provides. It is also hard: endoscopic scenes carry weak texture, non-Lambertian tissue, specularities, blood, smoke, deformable anatomy, rapid viewpoint change and tool occlusion [18], [19]. Modern estimators [11]–[15] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods build temporal reasoning into the model [1]–[4], at the price of a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting: *can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?* Backbone and temporal logic can then evolve independently, a new estimator replacing the previous one without the module needing its cost volume, hidden state, confidence head or training pipeline. We keep one inductive bias from the video-stereo literature — decide whether motion-aligned history should influence the current estimate, and constrain the correction to a convex interpolation between the two — and redesign everything around it for a black-box interface.

Does the learned head add anything simple temporal averaging would not? Matched fixed fusion, EMA, warped-memory propagation and forward–backward confidence weighting recover only a small fraction of TETHER's gain, while a ground-truth oracle shows the aligned memory carries far more than the policy exploits. The core problem is therefore not obtaining temporal evidence but deciding how much of it to use — and Sec. VI shows the head decides *how much* well while its own weight fails to rank *where* on data it was not tuned against, which is what leaves the oracle gap open. A second question is how such an intervention should be evaluated: smoothness is not correctness, and lower mean EPE hides whether refinement introduces new bad pixels or rare catastrophic frames, so we evaluate relative to the frozen prediction and separate geometry, temporal fidelity, benefit, harm and tail behaviour — and, because what a robot consumes is a map, carry the same intervention into an accumulated reconstruction and measure it there.

Our contributions are three.

(i) **One module, every backbone, every domain.** We measure a single frozen checkpoint across the full backbone $\times$ domain matrix — five estimators, two excluded from its training, over development, held-out and out-of-domain splits, including the joint unseen-backbone, unseen-domain cell usually left unmeasured — and it improves all fifteen cells, with the unseen estimators we cannot separate from the seen ones.

(ii) **A black-box interface that makes that possible.** We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a $154{,}874$-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 38-channel external evidence space, support-confined convex fusion and bounded corrected-state recurrence. Under a matched state machine the gain is not temporal smoothing.

(iii) **A protocol that scores the intervention, not the**

prediction. We measure introduced error alongside recovered error under fixed, prediction-independent support, and it changes conclusions: on development, every spatially uniform blend recovering a meaningful share of the gain buys it by *raising* the bad-pixel rate, while the one unlearned rule that varies in space does not. That trade is a property of the split as much as of the policy: out of domain, where the raw prediction is weak, none of the uniform blends pays in bad pixels either. Applied to our own claims the protocol is equally unkind: degenerate history-replay wins the no-reference metric, the head's weight inverts its relation to harm across splits, and what a map fuses improves on one split while degrading on the other, for a reason we can name.

II. Related Work

a) Deep stereo matching.: Stereo has moved from cost-volume regression to iterative recurrent estimation [11], [12] and broad-domain zero-shot models [13]–[15]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD introduced the reset/fusion mechanism we build on [1]; DynamicStereo [2], PPMStereo [4] and TC-Stereo [6] follow, all co-designed with the estimator, so a new backbone means a new temporal model. Match Stereo Videos exposes BiDAStabilizer as a plug-in on a frozen estimator [3], but propagates bidirectionally. A parallel line enforces consistency on monocular video depth by diffusion [24] or stabiliser networks [16]: a different output space, none causal at clip level. A further line adapts the stereo weights online [25], the opposite of our constraint.

c) What can actually be compared.: We are aware of no published method that is simultaneously causal, black-box and estimator-agnostic. CODD [1], whose two-gate decision we build on, is causal but not black-box: its fusion network consumes matching features from its own stereo network, which a cached third-party backbone does not expose. The 142-channel configuration of Sec. V-J began as the closest reproduction of it our interface admits — its projection replaced by our warp, a frozen ResNet-18 standing in for the features it would have taken from the estimator — so the comparison against it in Table IV is the nearest thing to running CODD here. Among black-box candidates StereoDiffusion [7] ships no implementation and [8], [9] publish code whose weights are no longer retrievable; the supplement audits the set. Two methods are runnable and we report both: BiDAStabilizer [3], which refines a frozen estimator as we do but bidirectionally, and TC-Stereo [6], which propagates state by rigid reprojection from per-frame pose — a privileged input our interface does not assume but SCARED-C supplies, so we run it as its authors intended (Sec. V-I).

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures.

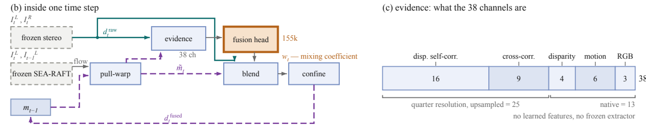

Fig. 1. **(b)** One time step; dashed grey is frozen, the single orange block is the only trained component and emits one scalar per pixel, w_t, rather than geometry. **(c)** The evidence space: none of the 38 channels carries ground truth, backbone identity or a future frame.

Recovered and introduced error must both be measured, which connects to selective prediction [17] and to the uncertainty literature [26]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [18] and DRENDS [22] supply the endoscopic stereo and time-of-flight geometry we evaluate on; D4D [21] enters only as the substrate for a no-reference control; SERV-CT [19] not at all, its pairs being non-consecutive. Downstream reconstruction and tracking pipelines consume exactly the per-frame geometry we refine [20], [27], which is why we treat refinement as an intervention on that geometry and, in Sec. V-H, measure it after accumulation rather than only before.

III. Method

A. External Causal Interface

A frozen estimator S produces positive-left disparity d_t^{raw} and its validity M_t^{raw}. The temporal module never reads the internal representation of S; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is $d_t^{\mathrm{fused}} = T_\theta(d_t^{\mathrm{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

B. Causal Motion Alignment

Frozen SEA-RAFT [5], [23] estimates both directions between the two already observed left images; the reverse direction is used only to measure reliability, and bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference. Previous geometry is pulled onto the current grid, $\widetilde{m}_t(x) = m_{t-1}(x + F_{t\to t-1}(x))$, by bilinear sampling with zero padding; warp support $\mathcal{S}_t$ records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way gives the forward–backward residual e_t^{FB} and the reliability cue $C_t^{FB} = \mathcal{S}_t \exp(-e_t^{FB}/\rho_t)$ with ρ_t scaled by the flow magnitudes. The zero padding is not a detail: it makes $\widetilde{m}_t$ exactly zero outside the source image, and Sec. V-K shows what happens when that value reaches the output.

C. Shared External Evidence

All estimators are mapped into one 38-channel tensor on the cache grid, built from raw disparity, aligned memory, the two images, the previous left image, the flow, C_t^{FB}, the warp support and the sampled historical validity. Twenty-five channels are disparity-geometry correlations formed at quarter resolution — 16 self-correlations of the two candidates and 9 raw-to-aligned cross-correlations over a dilated 3×3 neighbourhood — and 13 are formed on the grid directly: the two disparities, their signed and absolute residual, 6 motion components and 3 RGB. Normalisation is fixed before training. Nothing in this space is learned, no network evaluates it, and no ground truth, backbone identity or future frame enters it. An earlier configuration added 104 quarter-resolution channels derived from a frozen ResNet-18 [10] — 64 compressed feature maps and 40 confidence curves, supports and appearance correlations formed from them — for 142 in total. The channel ladder of Table IV reads off this decomposition: dropping only the 64 feature maps gives 78, dropping all 104 is TETHER's 38, and dropping its 25 disparity correlations leaves the 13 direct channels. Sec. V-J reports the 142-channel configuration as an ablation, because it costs 157,504 frozen parameters and three forward passes per frame and does not survive a three-seed comparison. Exact channel construction and constants are in the supplement.

D. Reset-and-Fusion Decision Head

The two-gate decomposition of the temporal decision, and the constraint that the correction be convex in the two candidates, are CODD's [1]; the architecture, the motion pathway, the evidence space, the supervision and the training are not.

A cache-grid pathway and a quarter-resolution context pathway (Fig. 1) feed a full-resolution reset branch predicting $r_t = \sigma(z_t^r)$ and a coarse fusion branch predicting $f_t = \uparrow_4 [\sigma(z_t^f)]$, giving the effective temporal weight $w_t = r_t f_t$ from 154,874 trainable parameters. Running the context branch coarsely is a compute choice on that branch, $16\times$ cheaper than full resolution, which Sec. V-J reports as an ablation. Layer widths are in the supplement.

Temporal evidence is only defined where the aligned memory is usable, so the per-pixel *support* is the conjunction of current raw validity, aligned temporal validity and warp support,

$$S_t = M_t^{\mathrm{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \tag{1}$$

and we confine the intervention to it. Final disparity is

$$\boxed{d_t^{\mathrm{fused}} = \begin{cases} (1-w_t)d_t^{\mathrm{raw}} + w_t \widetilde{m}_t, & S_t = 1, \\ d_t^{\mathrm{raw}}, & S_t = 0. \end{cases}} \tag{2}$$

Because $w_t \in [0,1]$ the fused disparity lies between the two candidates on the support, and off it the module is the exact identity on raw geometry. Applying (1) to the output and not only computing it is load-bearing (Sec. V-K). Reset output bias is initialized to -2 and fusion to zero, so initial intervention is conservative.

E. Bounded Corrected-State Recurrence

After refinement $m_t = d_t^{\mathrm{fused}}$, so alignment and fusion errors can become future evidence. We bound corrected-state lifetime: the first output of a sequence is raw, a re-anchor restores previous raw disparity as state, and otherwise the previous fused output is propagated.

The shipped policy permits four corrected-state uses before a raw re-anchor, so H denotes state lifetime rather than a temporal input window; at lifetime H the mean age of the state in use is $(H-1)/2$ frames, the quantity Sec. V-H finds the accumulated cost tracks. The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \to d_t^{\mathrm{fused}}$.

F. Training Objective

For target disparity g_t let $\Delta_t = |\widetilde{m}_t - g_t| - |d_t^{\mathrm{raw}} - g_t|$, so $\Delta_t < 0$ marks pixels where aligned memory is the better candidate; training support is the intersection of GT coverage, current validity, aligned temporal validity and warp support. The unweighted objective $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$ pairs a Huber term on the fused disparity with two decision terms supervising each branch against the sign of Δ_t at native margins 5 and 1 px, scaled to the cache grid, plus a weak pull of the fusion weight towards 0.5 where the two candidates are within the fusion margin. No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. Experimental Protocol

A. Training

The temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [18], using frozen predictions from S^2M^2-S [14], RAFT-Stereo [11] and Stereo Anywhere [13]; CREStereo [12] and Fast-FoundationStereo [15] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

Training uses D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs per backbone from three seen backbones, in non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0, selecting on development fused EPE. The budget is not a truncation: selection lands at epoch 8, 11 and 11 across three seeds, all inside the schedule, and a locked-recipe run to 47 epochs oscillates without trend from epoch 13 with no later epoch improving on it. One ablation (A4) diverges after epoch 10, which best-validation selection catches. Details in the supplement.

B. External Evaluation

DRENDS [22] supplies dynamic robotic-endoscopy stereo with calibration and metric geometry; we report five curated recordings, 7476 frames each, with all five estimators. Its disparity reference comes from metric ToF geometry through stereo calibration, so it is useful metric evidence but not independent stereo ground truth, and the ToF reference is itself temporally filtered. The zero-shot claim applies to the SCARED-trained checkpoint: no DRENDS data enters training, selection or calibration.

C. Unified Temporal-Refinement Evaluation

Evaluation support cannot depend on the prediction: $M_{\mathrm{eval}} = M_{\mathrm{GT}} \cap M_{\mathrm{protocol}}$, and invalid predictions on valid support are penalised rather than silently removed. Temporal fidelity is change error on fixed support, $\mathrm{DTCE} = |\Delta\mathrm{prediction} - \Delta\mathrm{target}|$ at lags $k \in \{1, 2, 4, 8\}$ (k=4 unless stated): it scores agreement with the *reference* change, not stillness — a frozen prediction has Δ=0 and is charged the whole of the reference motion, so the degenerate replay that wins the no-reference score of Sec. VII does not win this one. Intervention harm and benefit come from $\delta e = e^{\mathrm{fused}} - e^{\mathrm{raw}}$: harm and benefit masses $H^+ = \mathbb{E}[\max(\delta e, 0)]$, $B^+ = \mathbb{E}[\max(-\delta e, 0)]$ with their rates, newly broken and recovered bad pixels per threshold, and identity preservation. Aggregation is named at every number: macro is the mean of per-cell relative reductions, pooled sums before the ratio. The full family — spatial and metric-depth metrics, per-frame tails, risk–coverage [17], and the motion-compensated temporal proxies we do not build on — is defined in the supplement.

V. Results

The claim is one module across backbones and domains, so the results are ordered that way: the transfer matrix, the seed evidence that it is stable, why it works, what it does to a map, and what it costs.

A. Cross-Backbone Development Behaviour

$H = 4$ improves all five estimators in mean D2 EPE (Table I), from 1.41% to 8.19% averaging per-sequence reductions — 1.70% to 8.04% from the pooled cells the table prints — and all six unseen-backbone sequence groups improve; on D7 the mean excluding Stereo Anywhere is 5.15%. One D2 cell worsens, $\mathrm{S^2M^2}$-S on keyframe 2, by 0.28%: the first of three sequence-level exceptions in this paper, the others being Vid13 (Sec. V-I) and the development map (Sec. V-H).

B. Held-Out Cross-Backbone Confirmation

We next evaluate the already frozen $H = 4$ policy on held-out D7. All five backbones improve, including both unseen during temporal-head training: mean EPE decreases $0.4921 \rightarrow 0.4651$, or 5.49% pooling before the ratio and 5.40% averaging per-cell reductions.

The improvement is not restricted to the mean: Bad1, RMSE and P99 decrease for every backbone on both splits, so the EPE gain is not bought by a worse bad-pixel rate or a broader tail. New bad pixels do appear and are measured below. InvalidRate stays zero in all ten SCARED-C backbone–split evaluations, a property of SCARED-C and not of the module (Sec. V-K). The joint unseen-backbone, unseen-domain cell is the lower-right block of Table I.

C. External Zero-Shot Geometry on DRENDS

We evaluate the frozen checkpoint on five DRENDS recordings with all five estimators, two excluded from temporal-head training, 7476 frames each, nothing adapted or tuned for the domain. Disparity EPE, Bad3, metric depth MAE and RMSE improve in *every* one of the 25 recording–backbone cells, by -5.40, -19.51, -4.02 and -3.47% on average. Residual variance is set more by the recording than the estimator but not uniformly: the EPE change spans 0.87 points across backbones on Vid10 against 3.29 on Vid13.

Averaged over backbones the mean EPE reduction is 3.96% seen against 2.67% unseen on D2, 5.80 and 4.79% on D7, 5.57 and 5.16% on DRENDS: every combination positive. We can separate neither group from the other and report the margins rather than a test — with three seen and two unseen backbones and per-backbone margins from a single seed, absence of a detectable gap is not evidence of no gap. We read no ordering into the numbers either: on D2 the seen advantage is carried entirely by Stereo Anywhere and reverses without it, so we withdraw the claim we drew about which axis costs more. This establishes zero-shot geometric transfer under a ToF-derived reference; it does not establish universal robustness, and DRENDS remains a non-independent reference.

D. Seed Variance, and What It Reveals About Selection

We pre-registered a seed study before any result existed: only the seed varies, resampling is at sequence granularity rather than pixel, and selecting the best seed is prohibited.

Pooled over five backbones the relative Bad1 reduction is $12.73 \pm 0.32\%$ on D2 and $5.84 \pm 0.22\%$ on held-out D7; EPE gives $4.13 \pm 0.38\%$ and $5.61 \pm 0.42\%$, where $\pm$ is the standard deviation over three seeds throughout this paper. A paired sequence-level bootstrap is positive in every resample but the cells are not independent draws, so we report it as descriptive only (supplement); the distribution-free statement the splits can carry is that the effect is positive in $3/3$ and $4/4$ sequences, a sign test at $p = 0.125$ and $p = 0.0625$. No split here reaches $p < 0.05$ and none can at these sequence counts: we claim consistency of direction, never significance.

The spread is the strongest argument for the shipped configuration, and not what we found first. The 142-channel ablation is stable held out (5.15 ± 0.34 Bad1) and unstable on development, its seeds falling monotonically over a 5.03-point range. We read that as selection optimism, since checkpoints are selected on D2. Removing the learned evidence removes the instability instead — 0.32 points of D2 seed deviation against 2.53, with no change to the selection rule — so the instability was the evidence space and not the protocol.

E. Does the Learned Temporal Policy Matter?

We isolate the learned decision policy on D2, seed 0. All variants share the frozen predictions, the alignment, the support contract and the evaluation code, and all run unconfined so the comparison shares one implementation (confinement moves D2 EPE by 1.2×10^{-5} px). Figures are

TABLE I

ONE FROZEN TETHER CHECKPOINT, SEED 0, ACROSS FIVE ESTIMATORS AND THREE ARENAS; EVERY EPE AND BAD1 CELL IS RAW→TETHER, LOWER IS BETTER. TWO ESTIMATORS WERE EXCLUDED FROM TEMPORAL-HEAD TRAINING, SO THE LOWER-RIGHT BLOCK IS THE JOINT SHIFT USUALLY LEFT UNMEASURED. DTCE IS THE MACRO REDUCTION IN TEMPORAL CHANGE ERROR AT LAG k=4, HIGHER IS BETTER; ON DEVELOPMENT IT IS BACKBONE-DEPENDENT AND GIVEN IN SEC. V-F RATHER THAN HERE. CELLS ARE POOLED PER BACKBONE; PROSE FIGURES THAT AVERAGE PER BACKBONE–SEQUENCE CELL WILL NOT REPRODUCE FROM THIS TABLE, AND THE DRENDS BAD3 OF THE ABSTRACT IS NOT TABULATED. [T3.2]

		D2 development		D7 held out			DRENDS, out of domain		
Backbone	Status	EPE	Bad1 (%)	EPE	Bad1 (%)	DTCE	EPE	Bad1 (%)	DTCE
S^2M^2-S	seen	.2704 → **.2658**	1.39 → **1.27**	.5176 → **.4992**	6.44 → **6.03**	6.5	1.3477 → **1.2812**	62.55 → **58.98**	11.3
RAFT-Stereo	seen	.2691 → **.2623**	1.60 → **1.43**	.4671 → **.4296**	6.09 → **5.75**	7.3	1.3125 → **1.2371**	58.83 → **55.19**	8.5
Stereo Anywhere	seen	.3022 → **.2779**	2.18 → **1.74**	.4774 → **.4408**	6.23 → **5.77**	7.9	1.3685 → **1.2834**	62.73 → **57.99**	13.9
CREStereo	unseen	.2668 → **.2583**	1.57 → **1.40**	.5165 → **.4912**	6.35 → **6.05**	6.4	1.3027 → **1.2296**	55.17 → **52.21**	11.8
Fast-FoundationStereo	unseen	.2725 → **.2658**	1.60 → **1.44**	.4819 → **.4645**	5.92 → **5.65**	4.3	1.2838 → **1.2228**	54.88 → **52.06**	9.6

TABLE II

MATCHED-POLICY CLOSURE ON D2, SEED 0, ALL FIFTEEN BACKBONE–SEQUENCE CELLS POOLED BEFORE TAKING EACH RATIO; REDUCTIONS IN %, HIGHER IS BETTER, DTCE AT LAG k=4. EVERY ROW SHARES THE FROZEN PREDICTIONS, THE ALIGNMENT, THE SUPPORT CONTRACT, THE H=4 STATE MACHINE AND THE EVALUATION CODE — ONLY THE POLICY CHANGES. THE FIRST SIX ROWS LOAD NO CHECKPOINT; UNBOUNDED RECURRENCE IS OUR OWN TRAINED HEAD WITH THE RE-ANCHOR REMOVED. THE TUNED RESIDUAL RULE IS THE ONLY UNLEARNED POLICY THAT VARIES ITS WEIGHT IN SPACE, AND THE ONLY ONE THAT DOES NOT PAY FOR ITS MEAN GAIN IN BAD PIXELS. TETHER'S ROW IS SEED 0, THE WEAKEST OF ITS THREE ON THIS QUANTITY (3.69 AGAINST 4.13 ± 0.38), SO EVERY MARGIN HERE UNDERSTATES IT.

Policy	EPE	Bad1	DTCE$_{k=4}$	NewBad1
Fixed w=0.5 (≡ EMA2)	0.59	−2.64	6.26	0.214
EMA, three frames	0.30	−8.88	4.88	0.379
FB-confidence as the weight	−2.17	−32.35	−7.10	0.831
Warped raw previous, H=1	−0.03	−6.39	−4.33	0.373
Tuned residual rule	0.87	0.33	2.88	0.088
Aligned-memory propagation	−3.56	−41.70	−13.55	1.072
Unbounded corrected recurrence	−19.17	−143.48	−28.13	2.858
TETHER, H=4	**3.69**	**12.84**	**4.99**	**0.063**
Raw-vs-memory oracle (ceiling)	12.74	21.87	10.01	0.000

pooled over the fifteen backbone–sequence cells, whose raw EPE is 0.276202.

The fifteen policies (Table II, full grid in the supplement) are the horizon sweep and continuous recurrence for the learned head, fixed $w \in \{0.1, 0.2, 0.3, 0.5\}$ rising monotonically to 0.59%, EMA over two and three frames, forward–backward confidence as the weight, aligned-memory propagation, warped raw previous at H=1, and a tuned two-parameter rule on the raw-versus-memory residual swept over seventeen (α, τ) points. Per backbone, against whichever fixed or EMA policy is best *for that backbone*, the ratio ranges 3.9× to 9.9×: the absolute gain depends on the estimator, the margin over smoothing does not.

Three rows carry more than the mean. Forward–backward confidence is the informative failure — one of TETHER's own cues, yet used directly as the weight it worsens EPE by 2.17% and Bad1 by 32.35%. The uniform smoothing policies buy mean error with bad pixels, the more so the harder they blend: w=0.3 costs 0.18% more bad pixels for 0.49% of EPE, the best fixed blend 2.64% for 0.59%, EMA3 8.88% for 0.30%. The tuned residual rule limits that statement and we report it as such: it recovers roughly a quarter of our gain and does *not* pay in bad pixels. So the trade is not a property of blending but of blending *uniformly* — the moment a rule varies in space it stops buying mean error with tails. That is the closest an unlearned policy comes, and where the distance shows: 0.87% against 3.69% of EPE, breaking more pixels doing less. What the learned head adds is not varying in space; it is knowing where.

The closure is development work, and a reviewer is right to ask whether the margin survives the split. We repeat it out of domain, on the arena the freeze permits: on DRENDS with RAFT-Stereo, the best unlearned policy recovers 0.63% of EPE against our 5.80%, a ratio of 9× against 6× on D2. The denominator barely moves between the two arenas — uniform smoothing recovers the same small amount wherever it is run — so the widening is entirely ours, and it tracks how much raw error there is to recover (0.276 px of raw EPE on D2 against 1.286 out of domain, both on the closure's own support — Table I's protocol support and the BiDA common support of Sec. V-I give 1.31 and 1.20 for the same array), the same quantity Sec. V-H finds decides the accumulated benefit. Where the raw prediction is this weak the uniform blends no longer pay in bad pixels, so that trade is stated on D2 alone.

F. Why $H = 4$, and Why Recurrence Must Be Bounded

Recurrence must be bounded: with no re-anchor the intervention inverts, EPE degrading by 19.17% and Bad1 by 143.48% (Table II). Within the bounded range EPE rejects H=1, by 1.24 points, and does not separate the other three, whose spread is 0.21 (3.69, 3.73, 3.52%), while newly broken pixels rise monotonically from 0.052% to 0.128%. We froze H=4 before D7 opened on that risk–gain trade, and it is the last horizon at which change error also improves at lags $k \in \{2, 4, 8\}$ (4.99% at H=4 against −3.64% at H=6 for k=4). Two caveats. At k=1 DTCE improves at every horizon we tried, so the sign change belongs to the longer lags, not the metric. And on D2 the improvement is carried by one backbone and is *negative* for three: +14.0% for Stereo Anywhere and +0.8% for CREStereo against −0.3, −2.0 and −2.3%. A criterion whose sign one of five backbones decides cannot carry an operating point, so the horizon rests on breakage; on both confirmation splits DTCE is positive for every backbone (Table I). Sec. V-H sweeps the same horizon against accumulated geometry and finds the same risk structure from an independent measurement.

G. Refinement as an Intervention

Across both splits and all five backbones TETHER recovers 3.8–11.1× as many Bad1 pixels as it introduces, B^+/H^+ between 1.39 and 3.25, identity preservation above 99.85%. Per frame, 56% of the 15,535 held-out frames improve against 14% worsened (45% against 18% on development; the remaining third are exact Bad1 ties — cleaner frames whose sub-threshold corrections flip no pixel across the threshold), the worsened band hugging the diagonal while the improved mass extends far from it: the count-versus-magnitude asymmetry behind B^+/H^+.

TABLE III

ACCUMULATED GEOMETRY ON RAFT-STEREO: CHANGE IN WORLD-FRAME DISAGREEMENT, AS % OF THE EXCESS OVER THE GROUND-TRUTH FLOOR. POSITIVE IS BETTER. ALL ROWS SHARE THE POSES, THE ACCUMULATION AND THE FIXED SUPPORT; THE GRID IS COMPLETE (3 D2 AND 4 D7 CELLS PER POLICY). COPY-PREVIOUS IS THE STILLNESS CONTROL THAT WINS THE NO-REFERENCE SCORE OF SEC. VII; THE UNGATED WARP IS NOT DEGENERATE HERE BUT IS THE ALIGNMENT ABLATION — OUR WARP AT FULL WEIGHT, NO GATE.

Policy	D2 development	D7 held out
Copy previous (stillness control)	−54.0	+2.8
Ungated warp, full weight	−37.6	+7.2
TETHER, $H{=}1$	−3.6	+8.9
TETHER, $H{=}2$	−8.6	+10.4
TETHER, $H{=}4$ (shipped)	−15.5	+11.7
TETHER, $H{=}6$	−20.2	**+12.2**

H. What a Map Fuses, Not What a Frame Scores

Everything above scores a frame. What a robot consumes is a map, so we measure that too: SCARED-C's corrected poses let geometry be accumulated in one world frame and the disagreement between views of a point measured, reported as a percentage of the excess over the ground-truth floor. On held-out D7 refinement reduces it by 11.7%, and the sign is not carried by one estimator: across five backbones and four sequences all 20 cells improve, by +7.2 to +17.8%.

Our protocol requires any new measure to face the controls that defeat the no-reference score. Copy-previous, which removes 97.3% of that score, is here the *worst* policy in Table III: it degrades the map by 54.0% on D2 and returns 2.8% on D7 against our 11.7%. Disagreement between views is not reducible by repetition, because a replayed frame is wrong from the new viewpoint, so the measure survives the control our own contribution (iii) demands of temporal metrics.

The horizon sweep names the condition, on RAFT-Stereo across the complete grid; the five-backbone confirmation exists at the shipped horizon only (Sec. V-B), so the mechanism below is established on one estimator and its sign confirmed on five. On D2 the cost grows monotonically with state lifetime and decelerates — 3.6% at $H{=}1$, one warp and no accumulation, to 20.2% at $H{=}6$ — so a map pays not for the warp but for carrying it. The ungated warp at full weight costs 37.6% where the gated head at $H{=}1$ costs 3.6%, and on D7 already returns 7.2 of our 11.7: on a map most of the gain is alignment and most of the gate's value is damage avoided, the same division Sec. VI finds per pixel. Since alignment residual is a property of the motion while raw excess differs twofold between the splits (0.171–0.290 mm against 0.518–1.113), whether that residual is bought back is decided by the split.

Both splits are monotone in H and neither turns, so the horizon is set by an exchange rate rather than by a peak. Held-out benefit saturates while development cost keeps growing: successive steps buy +1.5, +1.3 and +0.5 points on D7 for −5.0, −6.9 and −4.7 on D2, so the price of lifetime rises from 3.3 to 9.4 points of development error per point of held-out gain. $H{=}6$ is therefore the best cell of the D7 column and the worst purchase on the sweep. We name this because the D7 column read alone would suggest a longer horizon, and it is the one place where reading a single column inverts the conclusion. $H{=}4$ was frozen on the per-frame risk–gain trade before D7 opened and before this measurement existed, and the map is the fourth criterion to place it there: of $H{=}4$ against $H{=}6$, frame EPE is unreadable at 0.21 points of spread, newly broken pixels rise, change error changes sign, and the map costs 4.7 to gain 0.5. A consumer that only accumulates may still prefer shorter — $H{=}1$ returns 76% of the held-out benefit at less than a quarter of the development cost, for 1.24 points of frame EPE — which is what makes the horizon a dial rather than an optimum.

I. Head-to-Head Against a Published Stabiliser

BiDAStabilizer [3] is the closest published plug-in stabiliser we could run. We compare on DRENDS, where neither method is in domain, on its own RAFT-Stereo *robust* prediction over all five recordings under one support neither influences (117.4M pixels). Two asymmetries run in opposite directions — the baseline is bidirectional, which our setting forbids; we are trained on surgical endoscopy, it is not — so we read the margin as an upper bound on our advantage, and an in-domain baseline remains the comparison we cannot run.

Raw EPE 1.199 becomes 1.175 under the baseline and **1.131** under TETHER; Bad1 44.92% becomes 44.81 and **43.39**, RMSE 1.876 becomes 1.776 and **1.710** — two refiners of a weak prediction (mean ground-truth disparity is 9.89 px), not a claim that either recovers a good one. TETHER is the only causal entry, at 62× fewer trained parameters. The exception is named: on Vid13 TETHER is behind the baseline and 0.013 px *worse than the raw prediction it refines*, the one recording of five where the intervention is a net harm. In domain the ordering is the same and wider (EPE 0.3029 → 0.2889, Bad1 2.09 → 1.80% against 0.3021 and 1.99% on that comparison's own support), measured where we are in domain and it is not.

The temporal measure reverses the ordering, and we report it for that reason: change error falls 64.9% for the bidirectional baseline against our 42.5% at lag 1 (37.2 against 13.8% at lag 4), in every recording at every lag. A method that sees the future is more temporally stable and simultaneously less accurate on every geometric metric here: a reader given only a consistency score would rank these two the wrong way round.

TC-Stereo [6], run as its authors specify with its rigid pose warp on SCARED-C's corrected poses (convention self-checked, supplement), is outside its domain before any temporal component runs (frame path EPE 0.524 against 0.303 raw on the same support); enabling the temporal path makes both metrics worse and raises change error at every lag. We read that as a co-designed architecture meeting a scene its rigid warp does not describe: a TC-Stereo trained on deforming tissue is an experiment nobody has run, ours included.

J. Ablating Our Own Design Decisions

Four design choices were ablated under a pre-registration naming held-out Bad1 as the endpoint and a three-seed spread of 0.37 points as the threshold below which a difference is not read — the standard deviation over the base configuration's three seeds of its pooled Bad1 reduction across all 35 backbone–sequence cells, computed before the registration was written; A4 remains a single run. A separate later registration governs *promotion* of a variant to the shipped model and decides that on development only, which is how the 38-channel head was promoted; the two rules are distinct and we apply each where it belongs.

TABLE IV

HELD-OUT RELATIVE REDUCTION (%), NOT ERROR: HIGHER IS BETTER, MACRO OVER BACKBONE–SEQUENCE CELLS. EACH INDENTED DEVIATION IS MEASURED AGAINST THE UN-INDENTED BASE DIRECTLY ABOVE IT; EVERY ROW'S PRE-REGISTRATION NAMES ITS BASE. DROPPING THE LEARNED EVIDENCE FROM THE 142-CHANNEL BASE IS WHAT TETHER NOW IS, SO THE SHIPPED MODEL HEADS THE TABLE AND THE CONFIGURATION IT WAS ABLATED FROM SITS BELOW. THE FULL-RESOLUTION AND UNCONSTRAINED-OUTPUT DEVIATIONS OF THE SHIPPED BASE ARE REPORTED IN THE TEXT. "3 SEEDS" ROWS ARE MEANS OVER THREE SEEDED TRAININGS; † ROWS ARE SINGLE RUNS.

Configuration	Evidence	D7 EPE red.	D7 Bad1 red.
TETHER (3 seeds)	38 ch	$\mathbf{5.61 \pm 0.42}$	$\mathbf{5.84 \pm 0.22}$
no FB cue in training†	38 ch	5.05	5.31
geometry only†	13 ch	4.44	4.64
+ learned evidence (3 seeds)	142 ch	5.06 ± 0.44	5.15 ± 0.34
no appearance†	78 ch	4.93	4.88
full-res context (3 seeds)	142 ch	5.24 ± 0.52	5.68 ± 0.16
unconstrained output (3 seeds)	142 ch	1.70 ± 1.43	6.97 ± 0.83

The result (Table IV) is not the one we expected, and it is why the shipped model is the smaller one. We began where the video-stereo literature does, with a frozen ResNet-18 [10] supplying appearance correlations and cost curves on top of the geometric channels — the configuration the four deviations were pre-registered against. Three sit above it on the held-out endpoint and the fourth, dropping appearance, sits inside the threshold: against the 142-channel base, none of the four registered decisions is load-bearing. Against the head we ship, two of the same decisions *are*: the quarter-resolution context loses 0.76 points when moved to full resolution (5.08 against 5.84 ± 0.22, one seed), and dropping the 25 correlations loses 1.20 (4.64, one seed). The learned evidence was absorbing the difference — with it present the architecture is insensitive to these choices, and removing it is what makes them matter. Removing the learned evidence entirely was then given the same three-seed treatment, under the promotion rule that decides on D2 with D7 reported afterwards, and it stays ahead on both endpoints with seed deviation eight times tighter (0.32 against 2.53 points of D2 Bad1). Out of domain the ordering holds: on the three backbones carrying three seeds of each, DRENDS EPE falls by $5.20 \pm 0.13\%$ against $4.83 \pm 0.15\%$. The learned evidence is not what makes the module work; it is what makes its training unstable, at 157,504 frozen parameters, three forward passes per frame and 13.8% of the runtime. Its apparent advantage on the development closure (4.05% against 3.69%) is a seed-0 artefact, and a sharp one: seed 0 is the *best* of three for that configuration and the *worst* of three for ours, so the pairing everyone would report is the worst available to us. Over three seeds the ordering reverses, $4.13 \pm 0.38\%$ against $3.55 \pm 0.45\%$.

Two rows need the registered rule read out loud. Full-resolution context wins the registered endpoint against the configuration it was registered against (5.68 ± 0.16 against 5.15 ± 0.34), an outcome the registration did not enumerate. On the base we ship, the same deviation loses 0.76 points; the registered configuration is also the better one.

Ablating convexity (A4) posts the largest held-out Bad1 reduction in the table, $6.97 \pm 0.83\%$, but its margin over us (1.13 points) is not separable from its own three-seed spread — four times ours, and twice the registered threshold, which assumes comparable spreads and cannot be applied here. Its EPE loss *is* separable: $1.70 \pm 1.43\%$ against our 5.61 ± 0.42, over four standard errors. We decline it on the metric where the difference is real rather than on the metric that suits us. The two trajectories put side by side say what the constraint buys: from development to held-out, A4's EPE gain falls $5.32 \rightarrow 1.70 \pm 1.43$ while ours rises $4.13 \rightarrow 5.61 \pm 0.42$, and the drop holds in all fifteen backbone–seed pairs — the unconstrained output buys in-domain accuracy that does not survive the split shift. The shipped-base twin repeats the pattern in one seed: it wins the registered endpoint there too (7.32 against 5.84 ± 0.22 Bad1), loses EPE (4.41 against 5.61 ± 0.42), and its development EPE margin (11.25 against our 3.69) does not survive the split — the escape hatch overfits development on either base. Its registered consequence we honour: the falsification clause required that if A4 matched or beat the canonical model held out, the claim attributing *cross-backbone* transfer to the convex action space be withdrawn rather than softened. It is withdrawn and appears nowhere in this paper; the cross-*split* statement above is a narrower claim, measured on A4's own three seeds and repeated by its twin.

K. The Support Contract Is Load-Bearing

Confining the output to (1), not merely computing it, is a single line. On DRENDS the difference is categorical: unmasked, the shipped head *regresses* depth RMSE in every one of the 25 recording–backbone cells, by $+221.9\%$ on average (range $+29.5$ to $+594.5$); confined, it improves in every one of the same 25 cells, by -3.5% (-6.7 to -1.8). Zero-padded out-of-bounds memory enters the blend as $\widetilde{m}_t = 0$ and bounded recurrence carries the collapse to the next re-anchor; the offenders are rare and bursty (300 of 38.8M pixels on one recording, aligned memory exactly 0.0 for every one) and, since invalid predictions on valid support are penalised rather than discarded, they are the entire source of a depth tail we had attributed to the external domain. On SCARED-C the same ablation is nearly inert: the contract costs a little in-domain gain, buys well-posed behaviour out of it, and is almost invisible to anyone validating in domain only.

VI. Analysis

Alignment is necessary but not sufficient: the aligned memory carries a 12.74% oracle ceiling that indiscriminate use does not collect (Table II). The mechanism is alignment plus learned graded intervention plus bounded recurrence, not smoothing; the head never solves stereo matching, it chooses between two candidates.

a) What each branch of $w_t = r_t f_t$ does.: Suppressing either factor at inference, the trained head unchanged, separates their jobs. Fusion alone keeps most of the gain, 3.49% against 3.69%, but breaks a third more pixels (0.084% New-Bad1 against 0.063%); reset alone keeps 2.23% and breaks 0.211%. The fusion weight sets how much is gained, the reset gate how little is broken, and a paper reporting only the product and only mean error would have seen neither. Sec. V-H reaches the same division from accumulated geometry.

Ablating convexity (A4) collapses held-out mean EPE gain to $1.70 \pm 1.43\%$ while improving RMSE and worst case: the failure is broad degradation of pixels already close, not occasional large error, so the constraint buys mean accuracy rather than safety from large corrections.

Two limits are worth stating. Against the raw-versus-memory oracle of Table II — measured on TETHER's own aligned memory, since under $H{=}4$ that memory is the previous fused output on three frames of four — TETHER recovers 29.0% of the available gain on D2 pooled and 25.2% per cell, so the open problem is deciding *where* to intervene.

And the effective weight w_t is an intervention magnitude rather than an uncertainty. Per pixel it ranks harmful intervention at AUROC 0.393–0.408 on D2 and 0.488–0.507 on held-out D7. The development figure is not chance-level but *inverted* — read in the benign direction it is 0.592–0.607, so a larger weight marks a pixel more likely to have been helped — while on held-out data the same reading gives 0.493–0.512 and orders nothing. The signal exists only

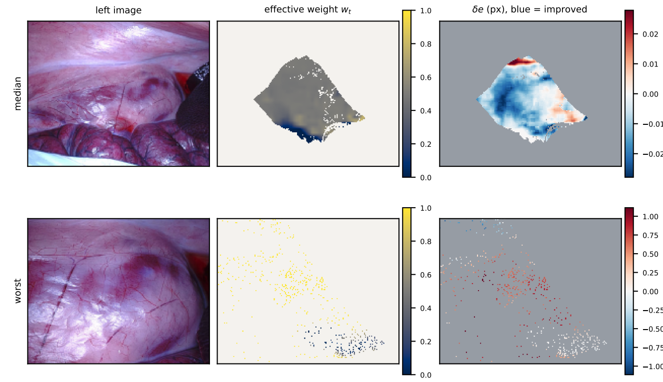

Fig. 2. Where the module intervenes, and whether it helped there. Grey and off-white are off support, where (2) makes the module the exact identity: w_t is undefined and δe is exactly zero, so those pixels are not drawn at the centre of the diverging scale. **Top:** the median frame of a D2 sequence; the module acts on 19.2% of pixels at a mean weight of 0.46 and improves 82.8% of what it touches. **Bottom:** the worst frame of the same sequence, included by rank rather than choice; it acts on 2.2% of pixels at a mean weight of 0.79 and improves 12.9% of them. The δe scales differ by $40\times$ between rows (±0.03 against ±1.11 px). "Acts on" is the fraction of the frame's pixels on support. The failure mode is few pixels, high weight, wrong — and it is a tail, not a trend: across the sequence's 1,861 frames with defined weight, per-frame mean weight correlates *negatively* with per-frame harm (Spearman -0.68), the benign direction the per-pixel ranking gives on development; on a held-out sequence the correlation weakens to -0.26 with the same sign.

on the split the checkpoint was selected on, which is the selection optimism of Sec. V-D appearing a second time, and does not survive to data the head was not tuned against. The 142-channel configuration has the same shape, so this belongs to the formulation and not the evidence space. It does not contradict the intervention ratios, which factorise into a count and a magnitude term needing no per-pixel ranking: a policy can be good on average and uninformative per pixel, which makes the oracle gap a ranking problem rather than a capacity one.

VII. Discussion and Limitations

a) What the splits support.: The closure is development work on D2: it supports mechanism analysis and the pre-frozen $H = 4$ operating point, not confirmation. One external domain does not establish universal robustness, DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth, and per-backbone D7 margins come from one seed — the three-seed study bounds only the pooled effect.

b) What the metrics cannot say.: No-reference temporal consistency is gameable, which we show on D4D [21], used only as the substrate for that control: the module reduces the no-reference score by 53.9%, copy-previous by 97.3%, and warped-previous scores exactly zero by construction. The accumulated-geometry measure of Sec. V-H survives those same controls, which is why the robotics claim rests on it rather than on consistency. Pull-warp support detects out-of-frame sampling but not internal disocclusion.

c) Where the intervention costs a map.: On development D2 the mapping measurement runs the other way: refinement *increases* world-frame disagreement by 15.5%

at the shipped horizon, uniformly, all 15 cells negative. Sec. V-H names the term responsible and shows it grows monotonically with state lifetime while the single warp costs almost nothing. The account is consistent with every cell we have but is not proven: the two splits' excess ranges never overlap, so no backbone makes D2 as hard as D7, and within D2 the per-backbone ordering does not follow excess either. What the grid establishes is that the reversal is a property of the split and of the horizon, not of one estimator. A module that refines a prediction already close to its reference can cost a map more than it returns, and the shorter the state lifetime the less it costs.

d) Cost, and where it is.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes: $28.90\,$ms overhead at $95.2\,$MB peak VRAM on an H100. The frozen estimator dominates, so the end-to-end rate is set by the backbone, $11.6\,$FPS on Fast-FoundationStereo (overhead 33.5%) to 1.4 on StereoAnywhere (4.0%); *no real-time claim is made*. Stage timing locates it: the two SEA-RAFT passes are 26.2 of the $28.9\,$ms, against 0.67 for the evidence tensor and 1.00 for the head; the tensor is that cheap because TETHER builds no learned features, where the 142-channel ablation spends $5.13\,$ms there. The reverse pass exists only to compute C_t^{FB}, and three measurements — each a single seed, the held-out ones under a pre-declared threshold — separate what the cue is for. Training without it costs 0.53 points of held-out Bad1 (5.31 against 5.84 ± 0.22), so it is load-bearing in training. Pinning it to a constant at inference costs the shipped checkpoint $+0.08$ held out (5.77 against that same seed's 5.69; the three-seed mean is 5.84 ± 0.22 — the comparison is within one checkpoint, which is the deployment question) — and the head trained *without* the cue is equally indifferent to receiving the real one (-0.02), so the inference indifference is architectural, not an accident of one checkpoint. The cue is a training-time scaffold: **train with it, deploy without the reverse pass**, taking the module from 28.9 to $\sim 15.8\,$ms on the unchanged checkpoint. We evaluate perception geometry and what a map accumulates from it, not downstream control, and demonstrate neither clinical safety nor patient benefit.

VIII. Conclusion

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: matched fixed and EMA blends under an identical state machine recover a sixth of it on development — paying in bad pixels there — and a ninth out of domain. One frozen $H = 4$ checkpoint improves all five estimators on held-out D7 and all 25 recording–backbone cells zero-shot on DRENDS.

Five results bound what may be claimed, and they are why the evaluation protocol is part of the contribution. No-reference consistency is won outright by replaying warped history, while the accumulated-geometry measure is not, which is why the robotics claim rests on the second. The fusion weight ranks harm only on the split it was selected on, so it is an intervention magnitude and not an uncertainty. Selection optimism on development is a property of the evidence space rather than of the protocol. A module that declares a support must apply it to what it emits. And what a map fuses improves held out on every backbone while degrading on development, the term responsible being accumulated alignment residual — so the horizon is an exchange rate rather than an optimum, one whose price rises threefold across the sweep and which we froze on frame evidence. What remains open is *where* to intervene.

OPEN BLOCKERS — DRAFT ONLY

This page renders only while `\internaldrafttrue` and costs nothing in the submission build. Tiers are ordered by consequence, not by position in the text. **T1 must be cleared.** T2 is what a reviewer will hit. T3 is cheap polish and can ship unresolved if time runs out. Clear an item by deleting its `\bitem` here *and* its inline `\B{}` tag.

Cleared this pass: T1.2 (the H=6 D7 map cell was n=3 against n=4 elsewhere) and T3.4 (two mapping cells missing) — both closed by the completed $35/35$ grid. Then T1.1 (the 0.37 threshold is the base configuration's three-seed sd of pooled Bad1 over all 35 cells, computed 2026-08-16 02:32, before the 09:40 registration — the arithmetic objection compared against the D7-only sd, a different quantity; now defined in the text) and T1.3 ("acts on" defined as support fraction; the full per-frame series over $1{,}861$ frames shows weight correlates *negatively* with harm, Spearman -0.68 — the worst frame is a tail, not a reversal, and the caption now says so). Development-only caveat on the latter: the series on a held-out sequence has not been run. Then T2.4 (support contract re-measured on the shipped head, paired over the same 25 DRENDS cells: unmasked $+221.9\%$ depth RMSE in $25/25$, confined -3.5% in $25/25$; the 142-channel unpaired version is replaced) and T2.5 (the FB-free training ran and *failed* the threshold: 5.31 against 5.84 ± 0.22 held-out Bad1, so the reverse pass is load-bearing at training time and the efficiency claim is withdrawn — the limitation paragraph now states both halves; one seed, conservative direction). Then T2.2, with a retraction inside it: a first timing of the full-res configuration (51.1 ms) was measured on a loaded node and is contaminated — stages identical between the two configurations inflated $4\times$ while the only stage the variant changes moved $1.01 \rightarrow 1.31$ ms, consistent with $16\times$ a ~ 0.02 ms branch. The passage now reads the comparison by the registered rule (indistinguishable in accuracy; the deviation buys nothing and spends $16\times$ the branch's compute) and no longer claims "ahead on both endpoints" across differences the threshold forbids reading. The paired quiet re-measurement landed and confirmed the bound exactly: shipped 29.72 ms against 35.65 for the full-resolution configuration, head 1.05 against 1.35 — the branch is $+0.30$ ms. This pass also closes T2.1 (the twins are measured: full-res loses 0.76 on the shipped base, the escape hatch wins the endpoint and loses EPE there too, and the two-base contrast — learned evidence absorbing the deviations — is now the section's finding), T2.3 (A4 at three seeds, A5 and A6 in the table, no stale single-run sentence remains), T2.5's completion (deploy-without stated with all three arms, each declared single-seed), and T3.3 (the missing third are exact Bad1 ties, said in V-G). T3.1 closes with the decomposition read from the builder: the 104 learned channels are 64 compressed feature maps plus 40 confidence curves, supports and appearance correlations formed from them, and the $142/78/38/13$ ladder now reconstructs in one clause of III-C. One checkpoint-baseline clarification in the runtime paragraph (5.69 is the same seed's own unpinned number, 5.84 ± 0.22 the three-seed mean) and the conclusion's out-of-domain fraction corrected to a ninth.

Tier 1 — factual. Do not submit with these open.

– *(No open Tier-1 items.)*

Tier 2 — a reviewer will hit these.

– *(No open Tier-2 items.)*

Tier 3 — cheap. Ship without if time runs out.

T3.2 **Prose numbers on D7 and DRENDS do not reconstruct from Table I** (they use per-cell aggregation, the table pools per backbone). D2 reconstructs exactly, so a reviewer who verifies D2 and then fails on D7 concludes something is wrong. The caption now says so; a per-sequence table in the supplement, pointed to here, would close it properly. Also: the DRENDS macro EPE (-5.40) is numerically identical to the D7 macro (-5.40) — coincidence, but print one more digit on one of them.

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